

double integral in polar coordinates

equations you will need to know

$$x^2 + y^2 = r^2$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\iint_A F(x,y) dA$$

$$\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} (r \cos \theta, r \sin \theta) r dr d\theta$$

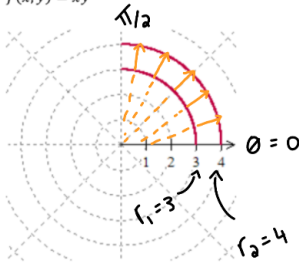
the first equation was changed to the second one

θ_1 must be smaller than θ_2

★ If you are dealing with a circle you need to use polar coordinates

3. Set up an iterated integral in the polar coordinates for the volume of the solid under the graph of the given function and above the region D . Evaluate the integral to find the volume of the solid.

a. $f(x,y) = xy^2$



• that is the area you are integrating (hard line)

$$\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} xy^2 r dr d\theta$$

replace $x = r \cos \theta$ and $y = r \sin \theta$

$$\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} (r \cos \theta) (r \sin \theta)^2 r dr d\theta$$

$$\int_0^{\pi/2} \int_3^4 r^4 \cos \theta \sin^2 \theta dr d\theta$$

$$\int_3^4 r^4 \cos \theta \sin^2 \theta dr \Rightarrow$$

$$\left[\frac{r^5}{5} \cos \theta \sin^2 \theta \right]_3^4$$

$$\int_0^{\pi/2} \frac{781}{5} \sin^2 \theta \cos \theta d\theta$$

$$u = \sin \theta$$

$$du = \cos \theta d\theta$$

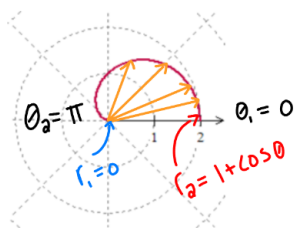
$$\frac{du}{\cos \theta} = d\theta$$

$$\frac{781}{5} \int_0^{\pi/3} (u)^2 \cos \theta \cdot \frac{du}{\cos \theta} \Rightarrow \frac{781}{5} \left[\frac{(\sin \theta)^3}{3} \right]_0^{\pi/3}$$

$$\frac{781}{5} \left(\left[\frac{(\sin(\pi/2))^3}{3} \right] - \left[\frac{(\sin(0))^3}{3} \right] \right)$$

$$\frac{781}{5} \left(\frac{1}{3} \right) = \boxed{\frac{781}{15}}$$

b. $f(x, y) = 1$ where $r = 1 + \cos \theta$



$$\int_0^{\theta_2} \int_{r_1}^{r_2} 1 \cdot r \, dr \, d\theta$$

$$\int_0^{\pi} \int_0^{1+\cos \theta} 1 \, dr \, d\theta$$

$$\int_0^{1+\cos \theta} 1 \cdot r \, dr \, d\theta \Rightarrow \left[\frac{r^2}{2} \right]_0^{1+\cos \theta} \Rightarrow \left[\frac{(1+\cos \theta)^2}{2} \right]$$

$$\frac{1}{2} \int_0^{\pi} 1 + 2\cos \theta + \cos^2 \theta \, d\theta \Rightarrow \frac{1}{2} \int_0^{\pi} 1 \, d\theta + \int_0^{\pi} 2\cos \theta \, d\theta + \int_0^{\pi} \cos^2 \theta \, d\theta$$

$$\frac{1}{2} \left[\theta + 2\sin \theta + \frac{1}{2}\theta + \frac{1}{4}\sin(2\theta) \right]_0^{\pi}$$

$$\frac{1}{2} \left[\pi + 2\sin(\pi) + \frac{1}{2}(\pi) + \frac{1}{4}\sin(2\pi) \right] - \left[0 + 2\sin(0) + \frac{1}{2}(0) + \frac{1}{4}\sin(2 \cdot 0) \right]$$

$$\frac{\pi}{2} + \sin(\pi) + \frac{\pi}{4} + \frac{1}{8}\sin(2\pi)$$

$$\boxed{\frac{5}{8}\pi + \sin(\pi) + \frac{1}{8}\sin(2\pi)}$$

Use the Polar Coordinates to Find the Volume of the Curve $z = \sqrt{x^2 + y^2}$ and above the ring $1 \leq x^2 + y^2 \leq 4$

$$x^2 + y^2 = r^2 \Rightarrow z = \sqrt{x^2 + y^2} \Rightarrow z = \sqrt{r^2} = z = r$$

★ When you are dealing with Circular Objects you need to use Polar Coordinates



If a Problem say "above the x-axis" or in "the first quadrant" We only cover that Part of the circle but here there isn't a restriction so we are going to cover the whole circle

$$\begin{aligned} \sqrt{x^2 + y^2} = \sqrt{4} &\rightarrow r = 2 \\ \sqrt{x^2 + y^2} = \sqrt{1} &\rightarrow r = 1 \end{aligned} \quad \cdot \int_{\theta=0}^{\theta=2\pi} \int_1^2 r \, r \, dr \, d\theta$$

$$\int_1^2 r^2 \, dr \Rightarrow \left[\frac{r^3}{3} \right]_1^2 = \left[\frac{(2)^3}{3} - \frac{(1)^3}{3} \right] = \frac{7}{3}$$

$$\int_{\theta=0}^{\theta=2\pi} \frac{7}{3} \, d\theta \Rightarrow \left[\frac{7}{3} \theta \right]_0^{2\pi} = \left[\frac{7}{3} (2\pi) \right] = \boxed{\frac{14\pi}{3}}$$

Application of double integral

equations/ explanations

the mass of the lamina in region D

$$\text{Mass} = m = \iint_D \rho(x,y) dA$$

Momentum of the lamina about the x-axis

$$m_x = \iint_D y \rho(x,y) dA$$

Momentum of the lamina about the y-axis is

$$m_y = \iint_D x \rho(x,y) dA$$

Coordinates of the center of mass are $(\bar{x}, \bar{y})$ (these are not vectors)

$$\bar{x} = \frac{m_y}{m} \quad \bar{y} = \frac{m_x}{m}$$

Momentum of inertia or second momentum about each of the x-axis is given by

$$I_x = \iint_D y^2 \rho(x,y) dA$$

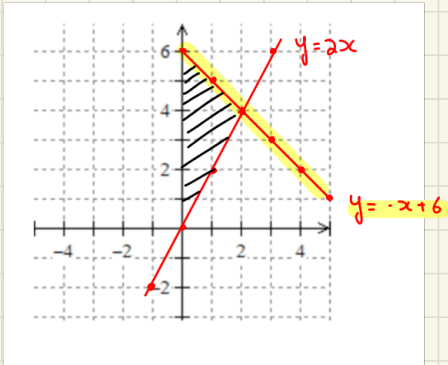
Momentum of inertia or second momentum about each y-axis

$$I_y = \iint_D x^2 \rho(x,y) dA$$

Momentum of inertia or second momentum about origin

$$I_o = \iint_D (x^2 + y^2) \rho(x,y) dA \quad \text{or} \quad I_o = I_x + I_y$$

given $P(x,y)=x$ and D is bounded by $y=2x$ and $y+x=6$, and y -axis



graph the slope of the two given equations

$$y=2x$$

$$y=-x+6$$

When you have variables they must be on the inner integral

$$\int_0^2 \int_{y=2x}^{y=-x+6} P(x,y) dy dx$$

We are integrating over the shaded area the x -axis is where the shaded area goes in the x -axis which is bounded from 0 to 2 it also goes from small to large the y -axis is going from $y=2x$ to $y=-x+6$ and it is going from bottom to top

What is the mass?

$$M = \int_0^2 \int_{y=2x}^{y=-x+6} P(x,y) dy dx \Rightarrow M = \int_0^2 \int_{y=2x}^{y=-x+6} x dy dx$$

$$\int_{y=2x}^{y=-x+6} x dy \Rightarrow [xy]_{y=2x}^{y=-x+6} \Rightarrow [x(-x+6) - x(2x)]$$

$$\int_0^2 -3x^2 + 6x dx \Rightarrow \left[-\frac{3x^3}{3} + \frac{6x^2}{2} \right]_0^2$$

$$[-x^3 + 3x^2]_0^2 \Rightarrow [-(2)^3 + 3(2)^2] = 4$$

Find the momentum of the particle about x-axis

$$M_x = \iint_0 y P(x,y) dA \Rightarrow \int_0^2 \int_{y=2x}^{y=-x+6} y x dy dx$$

$$\int_{y=2x}^{y=-x+6} xy dy \Rightarrow \left[\frac{xy^2}{2} \right]_{2x}^{-x+6}$$

$$\left[\frac{x(-x+6)^2}{2} - \frac{x(2x)^2}{2} \right] \Rightarrow \begin{aligned} (-x+6)(-x+6) &= x^2 - 6x - 6x + 36 \\ \frac{x(x^2 - 12x + 36)}{2} &= \frac{x^3}{2} - \frac{12x^2}{2} + \frac{36x}{2} \end{aligned}$$

$$\int_0^2 \frac{x^3}{2} - \frac{12x^2}{2} + \frac{36x}{2} - \frac{4x^3}{2} dx \Rightarrow \int_0^2 -\frac{3x^3}{2} - 6x^2 + 18x dx$$

$$\left[-\frac{3}{8}x^4 - 2x^3 + 9x^2 \right]_0^2$$

$$\left[-\frac{3}{8}(2)^4 - 2(2)^3 + 9(2)^2 \right] = \boxed{14}$$

Find the momentum of the particle about the y-axis

$$M_y = \iint_0 x P(x,y) dA \Rightarrow \int_0^2 \int_{y=2x}^{y=-x+6} x^2 dy dx$$

$$\int_{y=2x}^{y=-x+6} x^2 dy \Rightarrow \left[x^2 y \right]_{y=2x}^{y=-x+6}$$

$$\left[x^2(-x+6) - x^2(2x) \right] \Rightarrow \int_0^2 -x^3 + 6x^2 - 2x^3 dx$$

$$\int_0^2 -3x^3 + 6x^2 dx \Rightarrow \left[-\frac{3x^4}{4} + 2x^3 \right]_0^2$$

$$\left[-\frac{3(2)^4}{4} + 2(2)^3 \right] = \boxed{4}$$

Find the Center of Mass

$$\bar{x} = \frac{m_y}{m} = \frac{4}{4} = 1 \quad \bar{y} = \frac{m_x}{m} = \frac{14}{4} = \frac{7}{2}$$

$$\boxed{(1, \frac{7}{2})} \text{ this has to be in the Shaded region}$$

Find the momentum of inertia about the x-axis

$$I_x = \iint_D y^2 \rho(x, y) dA \Rightarrow \int_0^2 \int_{y=2x}^{y=-x+6} y^2 x dy dx$$

$$\int_{y=2x}^{y=-x+6} y^2 x dy \Rightarrow \left[\frac{xy^3}{3} \right]_{y=2x}^{y=-x+6}$$

$$\left[\frac{x(-x+6)^3}{3} - \frac{x(2x)^3}{3} \right] \Rightarrow \begin{matrix} (-x+6)(-x+6)(-x+6) \\ (-x+6)(x^2-12x+36) \end{matrix} \begin{matrix} x(-x^3+18x^2-108x+216) \\ = -\frac{x^4}{3} + \frac{18x^3}{3} - \frac{108x^2}{3} + \frac{216x}{3} \end{matrix}$$

$$-x^3+12x^2-36x+6x^2-72x+216$$

$$\int_0^2 -\frac{3x^4}{3} + \frac{18x^3}{3} - \frac{108x^2}{3} + \frac{216x}{3} dx \Rightarrow \left[-\frac{3x^5}{5} + \frac{18x^4}{4} - \frac{108x^3}{3} + \frac{216x^2}{2} \right]_0^2$$

$$= \boxed{\frac{264}{5}}$$

Find the moment of inertia about the y-axis

$$I_y = \iint_D x^2 \rho(x,y) dA \Rightarrow \int_0^2 \int_{y=2x}^{y=-x+6} x^3 dy dx$$

$$\int_{y=2x}^{y=-x+6} x^3 dy \Rightarrow \left[x^3 y \right]_{y=2x}^{y=-x+6}$$

$$\left[x^3(-x+6) - x^3(2x) \right] \Rightarrow \int_0^2 -3x^4 + 6x^3 dx$$

$$\left[-\frac{3x^5}{5} + \frac{6x^4}{4} \right]_0^2 \Rightarrow \boxed{\frac{24}{5}}$$

Find the moment of inertia about the origin

$$I_o = I_x + I_y = \frac{264}{5} + \frac{24}{5} = \boxed{\frac{288}{5}}$$

a lamina occupies the part of the disk $x^2 + y^2 \leq 1$ in the first quadrant. Find the center of mass if the density at any point is inversely proportional to its distance from the x -axis.

When two things are inversely proportional we say:

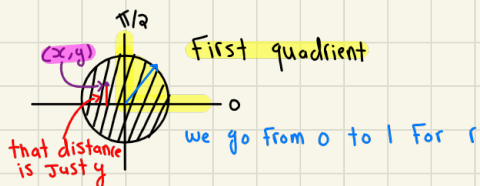
$$y \propto \frac{1}{x}$$

but to turn this into an equation, we multiply by a constant which can be any constant so:

$$y = \frac{k}{x}$$

also do notice that you were given a restriction which is the first quadrant

$$x^2 + y^2 \leq 1 \rightarrow r = 1$$



You were not given a $\rho(x, y)$ but you were told find the center of mass if the density at any point is inversely proportional to its distance from the x -axis. So at any point (x, y) the distance from that point is inversely proportional to that distance.

$$M = \iint_D \rho(x, y) dA \Rightarrow m = \int_0^{\pi/2} \int_0^1 \frac{k}{y} r dr d\theta$$

We would need to use polar coordinates because we are dealing with a circle

$$y = r \sin \theta$$

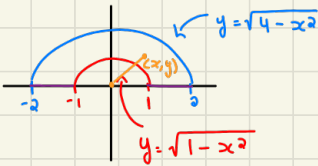
$$\int_0^{\pi/2} \int_0^1 \frac{k r}{r \sin \theta} dr d\theta \Rightarrow \int_0^{\pi/2} \int_0^1 \frac{k}{\sin \theta} dr d\theta$$

$$\int_0^{\pi/2} \int_0^1 \frac{k}{\sin \theta} dr d\theta \Rightarrow k \int_0^1 \csc \theta dr \Rightarrow k [r \csc \theta]_0^1$$

$$k \int_0^{\pi/2} \csc \theta d\theta \Rightarrow \left[\ln |\csc \theta - \cot \theta| \right]_0^{\pi/2}$$

$$M = \infty$$

a lamina consists of semi circles $\sqrt{1-x^2}$ and $y = \sqrt{4-x^2}$ and Portion of the x-axis that Joins to connect them Find its centers of mass if the density at any point is Proportional to distance from the origin



$$y = \sqrt{1-x^2}$$

$$y^2 = 1-x^2$$

$$y^2 + x^2 = 1$$

$$r = \pm 1$$

$$y = \sqrt{4-x^2}$$

$$y^2 = 4-x^2$$

$$y^2 + x^2 = 4$$

$$r = \pm 2$$

$P(x,y)$ = distance between (x,y) and $(0,0)$

$$P(x,y) = \sqrt{(x-0)^2 + (y-0)^2} = \sqrt{x^2 + y^2}$$

but it also mentioned that it was Proportional which just $A \propto B$ but in order to make it an equation we add a constant $A = kB$ the constant can be any letter so $P(x,y) = K\sqrt{x^2 + y^2}$

$$M = \iint_D P(x,y) dA = \int_{-2}^2 \int_{y=\sqrt{1-x^2}}^{y=\sqrt{4-x^2}} K\sqrt{x^2 + y^2} dy dx$$

~~*~~ We need to convert this to Polar coordinates because it's half a circle

We are going through the whole thing so from 0 to π

We are going from $r=1$ to $r=2$ even though we have a negative pair we will pick the positive

$$\star x^2 + y^2 = r^2$$

$$\int_0^\pi \int_1^2 k \sqrt{x^2 + y^2} r dr d\theta$$

$$\int_0^\pi \int_1^2 k \sqrt{r^2} r dr d\theta \Rightarrow \int_0^\pi \int_1^2 k r^2 dr d\theta \Rightarrow k \int_1^2 r^2 dr$$

$$k \left[\frac{r^3}{3} \right]_1^2 \Rightarrow k \left[\frac{8}{3} - \frac{1}{3} \right] = \frac{7k}{3}$$

$$\int_0^\pi \frac{7k}{3} d\theta \Rightarrow k \left[\frac{7}{3} \theta \right]_0^\pi \Rightarrow \boxed{\frac{7\pi k}{3}}$$